

ANGLO-CHINESE JUNIOR COLLEGE
DEPARTMENT OF CHEMISTRY
Promotional Examination

CHEMISTRY
Higher 2

9729

Question Booklet

06 October 2020
2 hours

Additional Materials: Multiple Choice Answer Sheet
Data Booklet
Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen **except** on the Multiple Choice Answer Sheet.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluids.

Sections A, B and C

Answer **all** questions.

Section A

There are **twenty five** questions in Section A. For each question there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Read the instructions on the answer sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section B and C

Questions for Section B are found in the Answer booklet. Answer them on the Answer Booklet.
Answer **all** questions in Section C on writing paper.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.

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ANGLO-CHINESE JUNIOR COLLEGE
Department of Chemistry

Section A: Multiple Choice Questions (25 marks)

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct and indicate your answers on the separate answer sheet provided.

- 1 An unknown compound **P** containing only carbon, hydrogen and oxygen was subjected to combustion analysis. 0.1 g of **P** on complete combustion gave 0.228 g of CO₂ and 0.0931 g of H₂O.

What is the molecular formula of **P**?

- A** C₂H₃O **B** C₃H₆O **C** C₃H₈O **D** C₂H₆O₂

Solution

Mass of C in P = Mass of C in CO₂ = (0.228)(12.0 ÷ 44.0) = 0.06218 g

Mass of H in P = Mass of H in H₂O = (0.0931)(2.0 ÷ 18.0) = 0.01034 g

Mass of O in P = 0.1 – 0.06218 – 0.01034 = 0.02747 g

	C	H	O
Mass / g	0.06218	0.01034	0.02747
Ar	12.0	1.0	16.0
Amount / mole	0.005182	0.01034	0.001717
Simplest Ratio	3	6	1

Empirical Formula of P is C₃H₆O

Molecular Formula of P is C_{3n}H_{6n}O_n where n is an integer (1, 2, etc.)

Best answer: C₃H₆O

- 2 12.50 cm³ of a 0.800 mol dm⁻³ solution of a metallic salt is oxidised exactly by 40.00 cm³ of 0.250 mol dm⁻³ acidified H₂O₂(aq).

If the original oxidation number of the metal in the salt was +2, what would be the new oxidation number?

- A** +3
B +4
C +5
D +6

Solution

Amount of salt = (0.01250)(0.800) = 0.01 mole

Amount of H₂O₂ = (0.04000)(0.250) = 0.01 mole



Amount of electrons given out by metal ion

= Amount of electrons taken in by H₂O₂

= 2 × 0.01

= 0.02 mole

0.01 mole of metal ion ≡ 0.02 mole of electrons

1 mole of metal ion ≡ 2 mole of electrons



- 3 The successive ionisation energies (IE) of two elements, **Q** and **R**, are given below.

IE / kJ mol ⁻¹	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
Q	1090	2350	4610	6220	37800	47000	-	-
R	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **Q** reacts with **R**?

- A** QR **B** Q₂R₃ **C** QR₄ **D** Q₄R

Solution

Q: largest increase from 4th IE to 5th IE.

Hence **Q** has 4 outermost electrons. **Q** is Group 14 element.

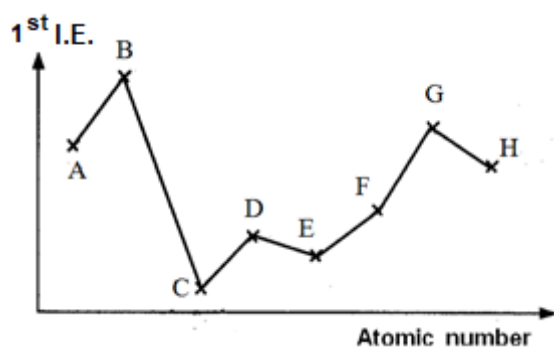
R: largest increase from 7th IE to 8th IE.

Hence **R** has 7 outermost electrons. **R** is Group 17 element.

Check from Data Booklet: **Q** is Carbon and **R** is Chlorine.

Hence compound is QR₄ or CCl₄.

- 4 The following graph shows the first ionisation energy of 8 consecutive elements from **A** to **H** in the periodic table, with atomic numbers between 3 and 20.



Which of the following statements is **incorrect**?

- A** The chloride of **E** has a giant ionic structure.
B The oxide of **F** has a giant molecular structure.
C The chloride of **D** conducts electricity in the molten state.
D Element **A** exists as discrete molecules at room temperature.

Solution

Atomic Number from 3 (Lithium) to 20 (Calcium).

Large decrease in 1st IE from **B** to **C**.

Either it is from Ne to Na, or Ar to K.

It cannot be from Ar to K, because if it is, then **E** is ²¹Sc, **F** is ²²Ti, etc.

Hence it is from Ne to Na.

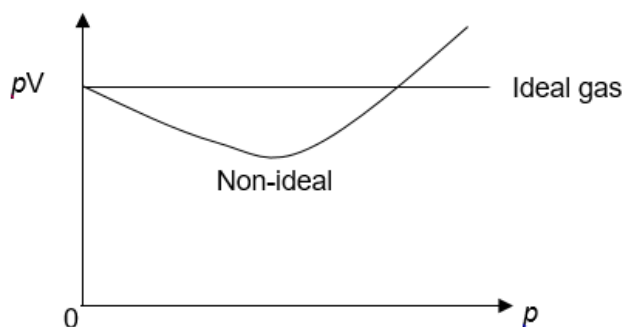
Option **A** is wrong, because AlCl₃ is a simple molecule.

Option **B** is correct, because SiO₂ has giant molecular structure.

Option **C** is correct because MgCl₂ is ionic compound that conducts electricity when molten.

Option **D** is correct because F₂ exists as discrete molecules at room temperature.

- 5 The value of pV is plotted against p for two gases, an ideal gas and a non-ideal gas, where p is the pressure and V is the volume of the gas.



Which gas shows the greatest deviation from ideality?

- A ammonia
- B methane
- C ethene
- D nitrogen

Solution

Assumption of an ideal gas: No intermolecular forces of attraction between molecules.

Ammonia has hydrogen bonds which are the strongest IMF of attraction among the four molecules. Hence it will show the greatest deviation from ideality.

Methane ($M_r = 15.0$) has instantaneous dipole – induced dipole which are the weakest IMF of attraction among the four molecules.

Both ethene ($M_r = 28.0$) and nitrogen ($M_r = 28.0$) also have ID – ID. These are stronger than that for methane, but weaker than that for ammonia.

- 6 Iodine and phosphorus each forms a trifluoride.

What are the shapes of these two molecules?

- | | <u>IF₃</u> | <u>PF₃</u> |
|---|-----------------------|-----------------------|
| A | trigonal planar | trigonal pyramidal |
| B | T-shaped | trigonal pyramidal |
| C | trigonal pyramidal | trigonal planar |
| D | T-shaped | trigonal planar |

Solution

Dot-Cross Diagram	Shape	Dot-Cross Diagram	Shape

- 7 Which of the following sets of compounds all contain the same type of forces of attraction between molecules of their own?

- 1 BF_3 , CH_4 , SF_6
- 2 AlCl_3 , I_2 , SiO_2
- 3 CO_2 , NO_2 , SiO_2

- A** 1 only
B 1 and 2 only
C 1 and 3 only
D 1, 2 and 3

Solution

BF_3 , CH_4 , SF_6 : All are simple molecules with ID – ID between molecules.

AlCl_3 and I_2 : Simple molecules with ID – ID between molecules.

SiO_2 : Giant molecule with covalent bonds between atoms.

CO_2 : Simple molecules with ID – ID between molecules.

NO_2 : Simple molecules with ID-ID and PD-PD between molecules.

SiO_2 : Giant molecule with covalent bonds between atoms.

- 8 Solid X has the following properties.

- It is insoluble in tetrachloromethane and water.
- It melts at 2730°C .
- It is a non-conductor of electricity.

Which of the following is a likely identity for X?

- | | |
|----------------------------|----------------------------|
| A Sodium | C Boron Nitride |
| B Boron trifluoride | D Rubidium chloride |

Solution

Sodium is soluble in water. Hence it cannot be X.

BF_3 is a simple molecule that has very low boiling point. Hence it cannot be X.

RbCl is an ionic compound that can dissolve in water. Hence it cannot be X.

Elimination of options left BN as the answer.

- 9 The use of the Data Booklet is relevant to this question.

A beaker containing 300 cm³ of water, at room temperature and pressure, was heated above a spirit lamp containing propan-1-ol (CH₃CH₂CH₂OH). After heating for 5 minutes, the temperature of the water increased by 14.0 °C, while the mass of the lamp decreased by 0.52 g. Assuming that heat transfer is 80% efficient, what is the standard enthalpy change of combustion of propan-1-ol?

- A -1621 kJ mol⁻¹
 B -2026 kJ mol⁻¹
 C -2532 kJ mol⁻¹
 D +2532 kJ mol⁻¹

Solution

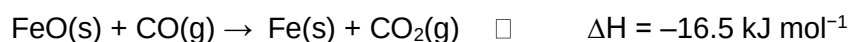
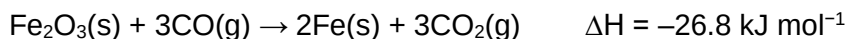
Heat absorbed by water = $mc\Delta T = (300)(4.18)(14) = 17,556 \text{ J}$

Heat given out by combustion = $17,556 \div 0.80 = 21,945 \text{ J}$

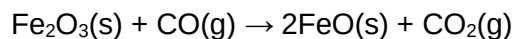
Amount of propan-1-ol = $0.52 \div 60.0 = 0.008667 \text{ mole}$

$\Delta H_c(\text{propan-1-ol}) = -21,945 \div 0.008667 = -2,532,018 \text{ J mol}^{-1} = -2,532 \text{ kJ mol}^{-1}$

- 10 The enthalpy changes of the following reactions are:

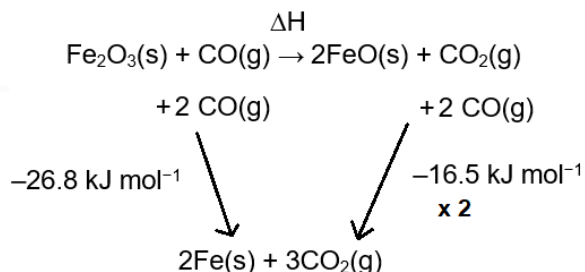


What is the enthalpy change for the reaction below?



- A +6.2 kJ mol⁻¹
 B -6.2 kJ mol⁻¹
 C -10.3 kJ mol⁻¹
 D -43.3 kJ mol⁻¹

Solution



By Hess's Law

$\Delta H = -26.8 - (-16.5)(2) = +6.2 \text{ kJ mol}^{-1}$

- 11** The experimental values of some enthalpy change terms are given in the table below.

Compound	NaBr(s)	KCl(s)
Lattice energy / kJ mol ⁻¹	-736	-698
Sum of enthalpies of hydration of separate ions / kJ mol ⁻¹	-754	-700

Which statement is correct?

- A** The difference in magnitude between experimental and theoretical lattice energies will be smaller for NaBr, compared to that for KCl.
- B** NaBr has more exothermic hydration energies than KCl, because the electronegativity difference between the ions in NaBr is greater than that in KCl.
- C** Upon dissolving, the ions in NaBr has stronger permanent dipole – permanent dipole attraction with water molecules, compared to those in KCl with water molecules.
- D** Enthalpy change of solution of solid NaBr is more exothermic than that of solid KCl.

Solution

Na⁺ is smaller than K⁺. Hence Na⁺ has greater polarising power than K⁺.
 Br⁻ is larger than Cl⁻. Hence Br⁻ is more polarisable than Cl⁻.
 Na⁺ will polarise Br⁻ to a larger extent compared to K⁺ polarising Cl⁻.
 Hence experimental LE will be greater for NaBr, compared to that for KCl.

The phrase “electronegativity difference between the ions” is wrong. Electronegativity does not apply to ions. It applies to two atoms in a covalent bond.

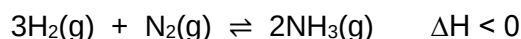
The phrase “the ions in NaBr has stronger permanent dipole – permanent dipole attraction with water molecules” is wrong. PD-PD does not apply to ions. It applies to simple molecules.

$$\Delta H_{\text{soln}} = -LE + \Delta H_{\text{hyd}}$$

$$\Delta H_{\text{soln}}(\text{NaBr}) = -(-736) + (-754) = -18 \text{ kJ mol}^{-1}$$

$$\Delta H_{\text{soln}}(\text{KCl}) = -(-698) + (-700) = -2 \text{ kJ mol}^{-1}$$

- 12** Haber process is an important reaction which allows the industrial synthesis of ammonia. Finely divided iron is used as a catalyst for this reaction.



Which of the following statements about this system are correct?

- The catalyst provides a surface for the reactant particles to react with each other.
 - The catalyst does not affect the equilibrium position.
 - For the same production rate of ammonia as before, the catalyst allows for a lower reaction temperature to be used.
- A** 1 only
- B** 1 and 2 only
- C** 1 and 3 only
- D** 1, 2 and 3

Solution

Option 1 is correct: H_2 and N_2 will be adsorbed onto iron surface. Bond breaking and formation will then occur.

Option 2 is correct: Catalyst increases forward and backward rates to the same extent. POE remains the same.

Option 3 is correct: Production Rate = Yield / Time. Catalyst does not change yield. However at the same temperature, catalyst will increase the rate of reaction. For the same rate of reaction as before, temperature needs to be decreased.

- 13** The rate law for the decomposition of a certain compound **Y** is $\text{rate} = k [\text{Y}]$.

At 25 °C, k is found to be $5.37 \times 10^{-3} \text{ s}^{-1}$. If the initial concentration of **Y** is $2.0 \times 10^{-2} \text{ mol dm}^{-3}$, what is the concentration of **Y** after 387 seconds?

- A** $1.07 \times 10^{-4} \text{ mol dm}^{-3}$
B $1.00 \times 10^{-2} \text{ mol dm}^{-3}$
C $5.00 \times 10^{-3} \text{ mol dm}^{-3}$
D $2.50 \times 10^{-3} \text{ mol dm}^{-3}$

Solution

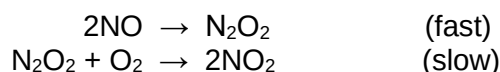
This is a first order reaction. This can also be inferred from the unit for rate constant.

$$k t_{1/2} = \ln 2$$

$$t_{1/2} = 129 \text{ sec}$$

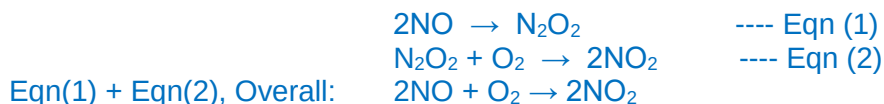
$$0.02 \xrightarrow{129 \text{ s}} 0.01 \xrightarrow{129 \text{ s}} 0.005 \xrightarrow{129 \text{ s}} 0.0025 = 387 \text{ sec}$$

- 14** It has been suggested that the mechanism of the reaction between NO and O_2 involves the following steps.



Which of the following statements are correct?

- 1 The stoichiometric equation is $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$.
 - 2 The rate equation is $\text{rate} = k [\text{N}_2\text{O}_2] [\text{O}_2]$.
 - 3 N_2O_2 is a catalyst.
- A** 1 only
B 1 and 2 only
C 1 and 3 only
D 1, 2 and 3

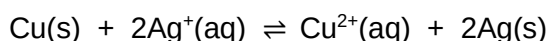
Solution

From slow step, Rate = $k [\text{N}_2\text{O}_2] [\text{O}_2]$

But N_2O_2 is intermediate. Rate equation must be expressed in terms of concentrations of reactants, not intermediate.

N_2O_2 is intermediate. It is produced in step 1, and used up in step 2.

- 15** Solid copper and silver ions react in an equilibrium reaction as follows.



Which of the following changes would shift the position of equilibrium to the right?

- 1 adding $\text{AgNO}_3(\text{aq})$
 - 2 adding water
 - 3 using powdered copper instead of a lump of copper of equal mass
- A** 1, 2 and 3
B 1 and 2 only
C 2 and 3 only
D 1 only

Solution

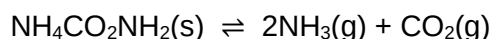
$[\text{Ag}^+(\text{aq})]$ increases. By LCP, POE shift to the right to decrease $[\text{Ag}^+(\text{aq})]$.

$$K_c = [\text{Cu}^{2+}(\text{aq})] \div [\text{Ag}^+(\text{aq})]^2 = \text{Amount of Cu}^{2+} \div (\text{Amount of Ag}^+)^2 \times \text{Volume}$$

Upon adding water, volume will increase. In order to keep K_c constant, the amount of Ag^+ needs to increase as well. Hence, POE shift to the left.

Powdered copper will increase rate as surface area of contact increases. But it does not affect POE.

- 16** Ammonium carbamate, $\text{NH}_4\text{CO}_2\text{NH}_2$, decomposes as follows:



Which of the following statements about this system in equilibrium is correct?

- A** Increasing the volume of the system will shift the equilibrium position to the right.
B Increasing the pressure of the system will decrease the equilibrium constant, K_p .
C Addition of a catalyst will increase the rate of the production and hence increase the equilibrium constant, K_c .
D Reducing the concentration of $\text{CO}_2(\text{g})$ causes the equilibrium position to shift to the left.

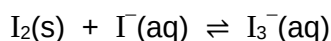
Solution

Assume that by increasing the volume, the pressure will decrease. By LCP, system will try to increase the pressure by shifting to the right to produce more NH_3 and CO_2 .
Option A is correct.

K will change only if temperature changes (provided that $\Delta H \neq 0$).
Hence options B and C are wrong.

When $[\text{CO}_2(\text{g})]$ decreases, by LCP, POE shift to the right.
Option D is wrong.

- 17** An equilibrium was reached after 1 mole of solid I_2 was added to 50 cm^3 of 3 mol dm^{-3} KI.



The reaction mixture was then rapidly cooled and filtered. Cold aqueous AgNO_3 was then added in excess to the cold filtrate. The mass of precipitate was 11.75 g.

Assuming that the equilibrium does not shift when temperature is low, calculate the value of K_c for the reaction.

- A** 2.22 **B** 2.00 **C** 0.15 **D** 0.11

Solution

Amount of KI = $(0.0500)(3) = 0.15 \text{ mole}$

Amount of AgI = $11.75 \div (107.9 + 126.9) = 0.0500 \text{ mole}$

At eqm, $[\text{I}^-(\text{aq})] = 0.0500 \div 0.050 = 1$

	$\text{I}_2(\text{s}) + \text{I}^-(\text{aq}) \rightleftharpoons \text{I}_3^-(\text{aq})$	
Initial / mol dm^{-3}	3	0
Change	-2	+2
Eqm	1	2

$$K_c = [\text{I}_3^-(\text{aq})] \div [\text{I}^-(\text{aq})] = 2 \div 1 = 2.00$$

- 18** The $\text{p}K_b$ values of four weak monoacidic bases are shown in the table below.

Base	J	K	L	M
$\text{p}K_b$	4.19	2.89	9.42	7.96

50.0 cm^3 of 0.1 mol dm^{-3} solutions of each base were prepared.

Which of the following statements is correct?

- A** The solutions need different amounts of acid to be completely neutralised.
B The pH of **J** will be higher than that of **L**.
C **K** is less basic than **M**.
D When water is added to 50.0 cm^3 of **L**, the $\text{p}K_b$ value of **L** increases.

Solution

Option A is wrong: All are monoacidic bases. For the same volume and concentration, they have the same amount of OH^- ions upon 100% ionisation. They need the same amounts of acid to be completely neutralised.

Option B is correct: J has lower pK_b value than L. Hence J is a stronger base than L. J will ionise more than L to give more OH^- ions.

Option C is wrong: K has lower pK_b value than M. Hence K is a stronger base than M.

Option D is wrong: K_b value (and hence pK_b value) depends only on temperature.

19 Which of the following could act as a buffer?

- 1 A mixture of 10 cm^3 of $0.10 \text{ mol dm}^{-3} \text{ NaHCO}_3(\text{aq})$ and 15 cm^3 of $0.10 \text{ mol dm}^{-3} \text{ Na}_2\text{CO}_3(\text{aq})$.
- 2 A mixture of 10 cm^3 of $0.10 \text{ mol dm}^{-3} \text{ NaOH}(\text{aq})$ and 15 cm^3 of $0.10 \text{ mol dm}^{-3} \text{ CH}_3\text{COOH}(\text{aq})$.
- 3 A mixture of 15 cm^3 of $0.10 \text{ mol dm}^{-3} \text{ NH}_3(\text{aq})$ and 20 cm^3 of $0.10 \text{ mol dm}^{-3} \text{ HCl}(\text{aq})$.

- A** 1 only
B 1 and 2 only
C 1 and 3 only
D 1, 2 and 3

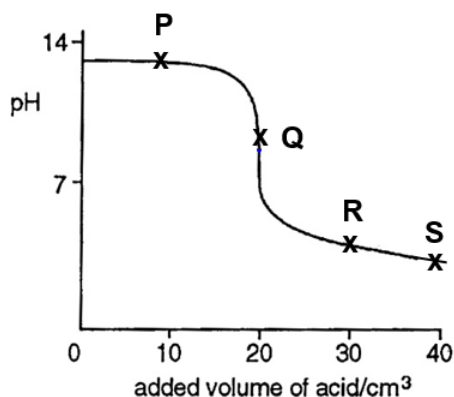
Solution

Option 1: This is an acid buffer solution of HCO_3^- (weak acid) and CO_3^{2-} (conjugate base).

Option 2: All OH^- will react with CH_3COOH (in excess). Resultant solution is CH_3COO^- (conjugate base) and CH_3COOH (weak acid). This is acid buffer.

Option 3: All NH_3 will react with H^+ (in excess). Resultant solution is NH_4^+ (weak acid) and H^+ (strong acid). This is not a buffer.

- 20 The graph below shows the change in pH when 0.10 mol dm^{-3} ethanoic acid is gradually added to 10 cm^3 of 0.20 mol dm^{-3} sodium hydroxide.



Which of the following points refers to the maximum buffer capacity?

- A Point P
- B Point Q
- C Point R
- D Point S

Solution

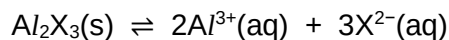
Point P: OH^- and CH_3COO^- (not a buffer)

Point Q: CH_3COO^- only (not a buffer)

Point R: CH_3COO^- and CH_3COOH . This is a buffer.
But $[\text{CH}_3\text{COO}^-(\text{aq})] > [\text{CH}_3\text{COOH}(\text{aq})]$. This is not MBC.

Point S: CH_3COO^- and CH_3COOH . This is a buffer.
Also $[\text{CH}_3\text{COO}^-(\text{aq})] = [\text{CH}_3\text{COOH}(\text{aq})]$. This is MBC.

- 21 A sparingly soluble aluminium salt dissociates in aqueous solution according to the equation below.



If the solubility product, K_{sp} of Al_2X_3 is S , what is the value of $[\text{Al}^{3+}(\text{aq})]$ at equilibrium?

- A $S^{\frac{1}{2}}$
- B $(2/5 S)^{\frac{1}{2}}$
- C $S^{\frac{1}{5}}$
- D $(8/27 S)^{\frac{1}{5}}$

Solution

$$K_{\text{sp}} = [\text{Al}^{3+}(\text{aq})]^2 [\text{X}^{2-}(\text{aq})]^3$$

$$S = (2y)^2 (3y)^3$$

$$y = \left(\frac{1}{108} S\right)^{\frac{1}{5}}$$

$$[\text{Al}^{3+}(\text{aq})] = 2y = \left(\frac{32}{108} S\right)^{\frac{1}{5}}$$

- 22 A solution containing $1.0 \times 10^{-5} \text{ mol dm}^{-3}$ of Ag^+ is slowly stirred into an equal volume of solution which contains Cl^- and Br^- ions, each with the same concentration of $2.0 \times 10^{-5} \text{ mol dm}^{-3}$. The numerical values of the solubility products are:

Species	K_{sp}
AgCl	1.8×10^{-10}
AgBr	3.3×10^{-13}

Which of the following statements is correct?

- A No precipitate will be seen.
 B Only AgBr precipitate will form.
 C Only AgCl precipitate will form.
 D A mixture of AgCl and AgBr precipitates will form.

Solution

$$\text{Ionic Product of AgCl} = [\text{Ag}^+(\text{aq})]^1 [\text{Cl}^-(\text{aq})]^1 = \left(\frac{1.0 \times 10^{-5}}{2}\right) \left(\frac{2.0 \times 10^{-5}}{2}\right) = 5.0 \times 10^{-11} \text{ mol}^2 \text{ dm}^{-6}$$

Since Ionic Product of AgCl $< K_{\text{sp}}$ of AgCl

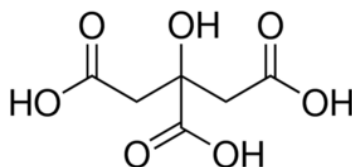
There is no AgCl precipitate.

$$\text{Ionic Product of AgBr} = [\text{Ag}^+(\text{aq})]^1 [\text{Br}^-(\text{aq})]^1 = 5.0 \times 10^{-11} \text{ mol}^2 \text{ dm}^{-6}$$

Since Ionic Product of AgBr $> K_{\text{sp}}$ of AgBr

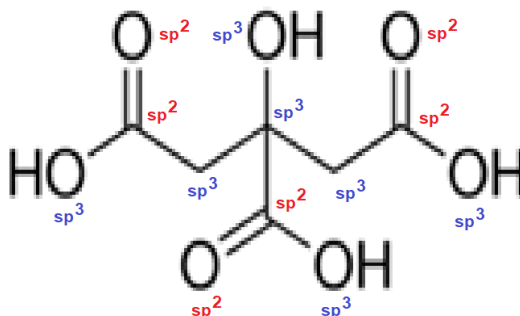
AgBr will precipitate out.

- 23 Citric acid is commonly found in citrus fruits such as lemon. It has the following structure:

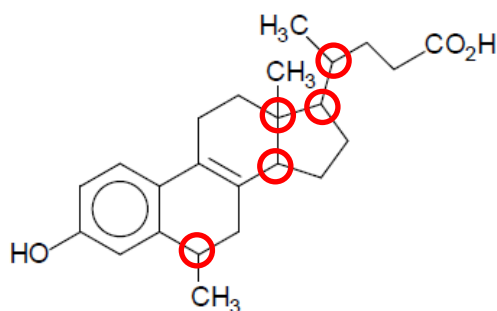


How many sp^2 and sp^3 hybridised atoms are there in a molecule of citric acid?

	<u>sp^2 atom</u>	<u>sp^3 atom</u>
A	3	4
B	3	7
C	6	4
D	6	7



24 How many enantiomers does this organic molecule have?

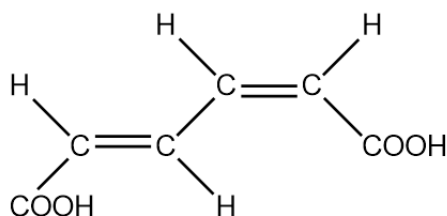


- A 8 B 16 **C 32** D 64

Solution

Max No. of enantiomers = $2^5 = 32$

25 Muconic acid has the following structure:



Which of the following statements about muconic acid is true?

- A** It exists in three stereoisomeric forms.
B It contains 2 π bonds.
C There are four sp^2 hybridised carbon atoms.
D It will exhibit intra-molecular hydrogen bonding.

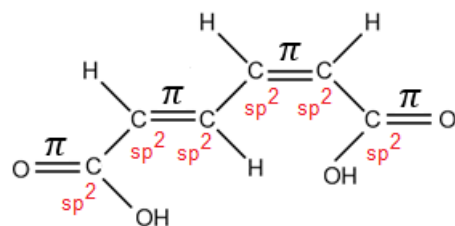
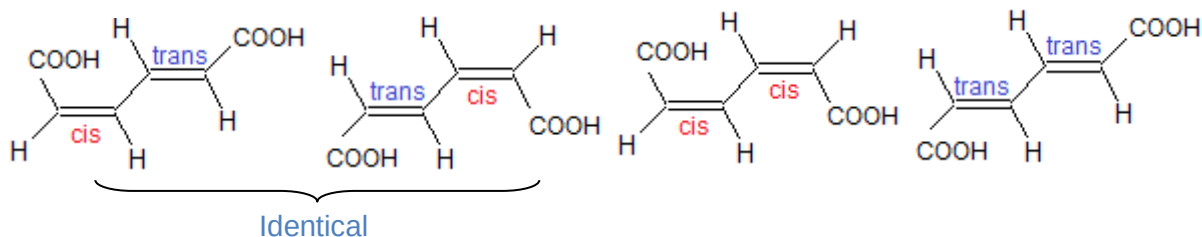
Solution

Options B is wrong: There are 4 π bonds.

Option C is wrong: There are six sp^2 hybridised carbon atoms.

Option D is wrong: The two $-\text{COOH}$ groups are too far apart to form hydrogen bonds.

Option A is correct:



1	B	6	B	11	D	16	A	21	D
2	B	7	A	12	D	17	B	22	B
3	C	8	C	13	D	18	B	23	D
4	A	9	C	14	A	19	B	24	C
5	A	10	A	15	D	20	D	25	A

Index Number ____	Name and Form Class 1_____	Tutorial Class 1CH_____	Subject Tutor
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ANGLO-CHINESE JUNIOR COLLEGE
DEPARTMENT OF CHEMISTRY
Promotional Examination

CHEMISTRY
Higher 2

9729

Answer Booklet

06 October 2020

Additional Materials: Multiple Choice Answer Sheet
Data Booklet
Question Booklet

2 hours

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen **except** on the Multiple Choice Answer Sheet.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Sections A, B and C

Answer **all** questions

Answer **all** questions in the spaces provided on the **Answer Booklet**. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use		
	Question Number	Marks
Section A (25)	1 – 25	/25
Section B (25)	1	/6
	2	/6
	3	/13
Section C (25)	1	/15
	2	/7
	3	/3
Presentation of answers		
Total: 75		

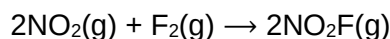


Section B (25 marks)

Answer **all** the questions in this section on the spaces provided.

- 1** Nitryl fluoride, NO_2F is a colourless gas which acts as a strong oxidising agent and fluorinating agent.

It is known that NO_2F can be produced from nitrogen dioxide and fluorine according to the equation below.



When NO_2 and excess F_2 were mixed together, $[\text{NO}_2]$ varied with time, as shown in the table below.

t / min	$[\text{NO}_2] / \text{mol dm}^{-3}$	$[\text{NO}_2\text{F}] / \text{mol dm}^{-3}$	$\frac{\Delta[\text{NO}_2\text{F}]}{\Delta t} / \text{mol dm}^{-3} \text{ min}^{-1}$
0.00	0.400	0.000	—
0.25	0.250	0.150	0.60
0.50	0.155	0.245	0.38
0.75	0.100	0.300	0.22
1.00	0.060	0.340	0.16
1.25	0.040	0.360	0.08
1.50	0.025	0.375	0.06

- (i) Explain why it is necessary to use excess fluorine in this experiment to confirm the order with respect to nitrogen dioxide.

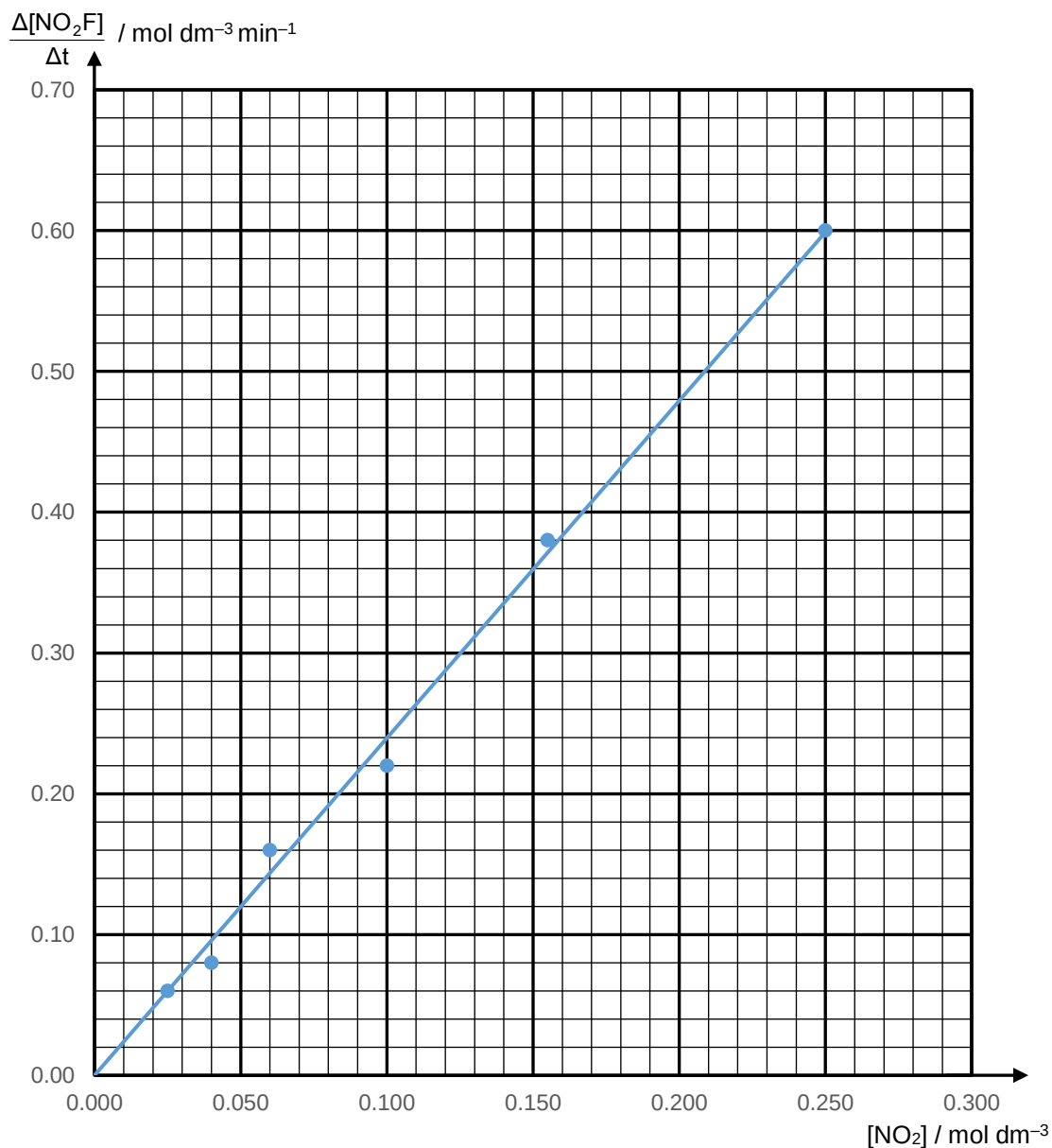
To keep the concentration of fluorine relatively constant so that the rate only depends
on the concentration of nitrogen dioxide/pseudo order with respect to nitrogen dioxide.

.....[1]

- (ii) The rate at a particular time can be approximated using rate = $\frac{\Delta[\text{NO}_2\text{F}]}{\Delta t}$.

Complete the table above by calculating $[\text{NO}_2\text{F}]$ produced (to 3 decimal places) and $\frac{\Delta[\text{NO}_2\text{F}]}{\Delta t}$ (to 2 decimal places) at the respective timings. [2]

- (iii) Plot a graph of $\frac{\Delta[\text{NO}_2\text{F}]}{\Delta t}$ against $[\text{NO}_2]$ on the given grid and use it to deduce the order of reaction with respect to nitrogen dioxide.



As a straight line graph with constant gradient passing through the origin is obtained,

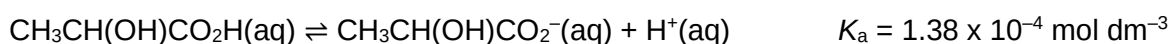
OR the rate is directly proportional to the concentration of NO_2 , hence the reaction is

first order

[3]

[Total: 6]

- 2 Lactic acid is a weak acid. It dissociates according to the equation shown.



- (a) (i) Lactic acid is found in some anti-wrinkle cosmetics. A particular sample analysed contains $0.0150 \text{ mol dm}^{-3}$ of lactic acid.

Calculate the pH of this sample. [2]

$$K_a = \frac{[\text{CH}_3\text{CH}(\text{OH})\text{CO}_2^-(\text{aq})][\text{H}^+(\text{aq})]}{[\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}(\text{aq})]}$$

$$1.38 \times 10^{-4} = \frac{[\text{H}^+]^2}{[0.0150]}$$

$$[\text{H}^+] = 0.00144 \text{ mol dm}^{-3}$$

$$\begin{aligned} \text{pH} &= -\lg[\text{H}^+] \\ &= -\lg(0.00144) \\ &= \underline{\underline{2.84}} \end{aligned}$$

- (ii) Buffered lactic acid is used in food confectionery.

A student prepared a similar buffered lactic acid solution for use in the laboratory. 25% of an original lactic acid solution was converted to its sodium salt with the addition of solid sodium hydroxide.

Calculate the pH of this buffered lactic acid solution. [1]

$$\begin{aligned} \text{pH} &= \text{p}K_a + \lg \frac{[\text{salt}]}{[\text{acid}]} \\ &= -\lg(1.38 \times 10^{-4}) + \lg \frac{25}{75} \\ &= \underline{\underline{3.38}} \end{aligned}$$

- (iii) Write an equation to show how pH change is resisted when small amounts of hydrochloric acid is added to the buffered lactic acid.



- (b) Calcium lactate is used in medicine as a calcium supplement, and in the food industry and water treatment as a coagulant.



Calcium lactate has a solubility of $0.0266 \text{ mol dm}^{-3}$ at room conditions.

- (i) Calculate the K_{sp} of calcium lactate. [1]

$$\begin{aligned} K_{\text{sp}} &= [\text{Ca}^{2+}][\text{CH}_3\text{CH}(\text{OH})\text{CO}_2^-]^2 \\ &= [0.0266][0.0532]^2 \\ &= \underline{7.53 \times 10^{-5}} \text{ mol}^3 \text{ dm}^{-9} \end{aligned}$$

- (ii) Calculate a value for ΔG at this temperature using $\Delta G = -RT \ln K$. Leave your answer in kJ mol^{-1} . [1]

$$\begin{aligned} \Delta G &= -RT \ln K_{\text{sp}} \\ &= -8.31(293) \ln(7.53 \times 10^{-5}) \\ &= \underline{+23.1} \text{ kJ mol}^{-1} \end{aligned}$$

[Total: 6]

- 3 Hennig Brand discovered phosphorus in 1669. It is the first element to be discovered that was not known since ancient times and is essential for life. Not surprisingly, Brand's original procedure involved heating urine until a white substance was obtained. Ammonium sodium hydrogen phosphate, $(\text{NH}_4)\text{NaHPO}_4$, is the main compound in urine which produced the white substance, sodium metaphosphate, NaPO_3 , on thermal decomposition. In the process, two gases are given off.

- (a) (i) Write a balanced equation for the thermal decomposition of $(\text{NH}_4)\text{NaHPO}_4$.



- (ii) Given a pure sample of solid $(\text{NH}_4)\text{NaHPO}_4$, briefly outline the experimental method to be performed to ensure that the thermal decomposition of $(\text{NH}_4)\text{NaHPO}_4$ is complete.
You do not need to specify quantities used.

1. Transfer a known mass of $(\text{NH}_4)\text{NaHPO}_4$ into a boiling tube/crucible.

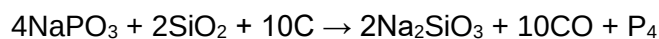
2. Gently heat the solid and then strongly.

3. Cool the boiling tube/crucible and weigh it for its contents.

4. Repeat steps 2 and 3 until a constant mass is obtained.

..... [2]

Phosphorus derives its name from Greek words which mean '*light-bringer*'. In later works, the white substance was heated with sand and graphite to liberate phosphorus which glowed in the dark.



- (b) (i) Use the information above and concepts from Atomic Structure to suggest how light energy is released from phosphorus molecules.

Thermal energy/heat cause electrons in phosphorus to be excited/promoted to a
higher energy level. De-excitation/relaxation of electrons back to the ground
state releases light energy.

..... [2]

- (ii) Explain, in terms of structure and bonding, why the melting point of silicon dioxide, SiO₂, is higher than carbon monoxide, CO.

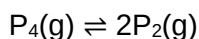
SiO₂ has a giant molecular structure while CO has a simple molecular structure.

The strong covalent bonds in SiO₂ require more energy to break than the weak

permanent dipole – permanent dipole forces of attraction between CO molecules.

[2]

Phosphorus exists in various allotropic forms and P₄ forms a vapour which dissociates into diatomic P₂ molecules at high temperatures.



When 0.883 g of P₄ was heated to 1200 °C in a 10.0 dm³ vessel and left to equilibrate, the total pressure at equilibrium measured was 0.124 atm.

- (c) (i) Calculate the initial pressure of P₄ in atm, assuming no dissociation has taken place yet. [2]

$$pV = nRT$$

$$p = \frac{\frac{0.883}{(31.0 \times 4)} \times 8.31 \times (1200 + 273)}{\frac{10.0}{1000}}$$

$$p = \underline{\underline{8716.5 \text{ Pa}}}$$

$$p = 8716.5 \text{ Pa} \div 101325 \text{ Pa atm}^{-1}$$

$$p = \underline{\underline{0.0860 \text{ atm}}}$$

- (ii) Hence, calculate the partial pressures of P₄ and P₂ at equilibrium. [1]
If you were unable to obtain a value in (c)(i), use 0.100 atm as the initial pressure of P₄, although this is not the correct value.

Pressure	$\text{P}_4(\text{g}) \rightleftharpoons 2\text{P}_2(\text{g})$	
Initial / atm	0.0860	0
Change / atm	-x	+2x
Equilibrium/ atm	0.0860 - x	2x

$$0.124 = 0.0860 - x + 2x$$

$$x = 0.0380 \text{ atm}$$

$$P_{\text{P}_4, \text{eqm}} = \underline{\underline{0.0480 \text{ atm}}}$$

$$P_{\text{P}_2, \text{eqm}} = \underline{\underline{0.0760 \text{ atm}}}$$

$$0.124 = 0.100 - x + 2x$$

$$x = 0.0240 \text{ atm}$$

$$P_{\text{P}_4, \text{eqm}} = \underline{\underline{0.0760 \text{ atm}}}$$

$$P_{\text{P}_2, \text{eqm}} = \underline{\underline{0.0480 \text{ atm}}}$$

(iii) Write an expression for K_p and include its unit.

[2]

$$K_p = \frac{P_{P_2}^2}{P_{P_4}} \text{ atm}$$

(iv) Calculate a value for K_p .

[1]

$$K_p = \frac{0.0760^2}{0.0480} \\ = \underline{0.120}$$

$$K_p = \frac{0.0480^2}{0.0760} \\ = \underline{0.0303}$$

[Total:13]

Section C (25 marks)

Answer **all** the questions in this section on the spaces provided.

1 Sulfur and phosphorus are both Period 3 elements.

(a) Generally, ionisation energies increase across a period.

(i) Explain why ionisation energies increase generally across a period. [2]

(ii) Explain why phosphorus has the higher first ionisation energy than sulfur instead. [1]

(i) There is an increasing nuclear charge due to increase in number of protons.

There is an approximately constant shielding effect as additional electrons are added to the same valence shell and provide negligible shielding effect.

Effective nuclear charge increases or valence electrons thus experiences stronger nuclear attraction.

(ii) Inter-electronic repulsion between paired electrons in the same orbital; less energy is required to remove the 3p electron from S than from P.

- (b) Sulfur forms a wide range of sulfides with phosphorus. One of the most well-known examples is the three-fold symmetric P_4S_3 used in strike-anywhere matches.



- (i) Suggest the bond angle about each sulfur atom in P_4S_3 . [1]
- (ii) The bond energies of the P—P and P—S bonds are 197 kJ mol^{-1} and 270 kJ mol^{-1} respectively.
Given the enthalpy change for the following transformation

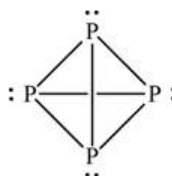
$$P_4S_3(g) \rightarrow P_4(g) + 3S(g) \quad \Delta H = +1029 \text{ kJ mol}^{-1},$$
show that there are six P—P bonds in P_4 . [2]
- (iii) Given that the four phosphorus atoms are identical in P_4 , draw the displayed formula of P_4 and state the shape about each atom. [2]
- (iv) P_4O_3 has a structure that is similar to P_4S_3 .
Explain why the O—P—O bond angle is smaller than the S—P—S bond angle. [2]

(i) 104.5° or 105°

(ii) $6(270) + 3(197) - 197x = 1029$

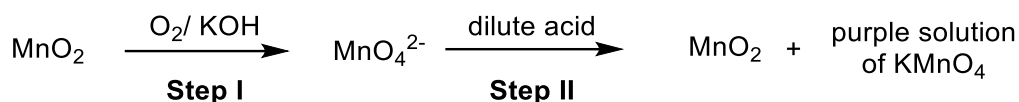
On solving, $x = 6$.

(iii) Trigonal pyramidal (wrt each P atom)



(iv) S is less electronegative than O; as a result, the electron density in the bond pairs are closer to the P atom in P_4S_3 than in P_4O_3 . There is less bond pair-bond pair repulsion in P_4O_3 .

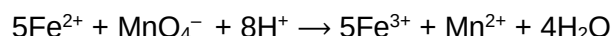
- (c) The reaction scheme below shows the reactions of some manganese compounds.



In **step II**, reaction of MnO_4^{2-} with dilute acid produced 0.174 g of MnO_2 and a purple solution of KMnO_4 . The KMnO_4 solution was found to react completely with 26.70 cm^3 of 0.75 mol dm^{-3} iron(II) sulfate in an acidic medium.

- (i) State the type of redox reaction in **step II**. [1]

- (ii) Given that the reaction between KMnO_4 and iron(II) sulfate is given as:



Show that MnO_2 and KMnO_4 are formed in the mole ratio 1:2 in **step II**. [2]

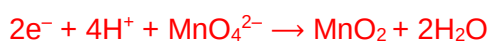
- (iii) Construct the balanced ionic equation for the reaction that occurred in **step II**. [2]

(i) Disproportionation.

$$(ii) n(\text{MnO}_2) = 0.174 / (54.9 + 16.0 \times 2) = 2.0023 \times 10^{-3}$$

$$n(\text{KMnO}_4) = (26.70 \times 0.75 \times 10^{-3}) / 5 = 4.005 \times 10^{-3}$$

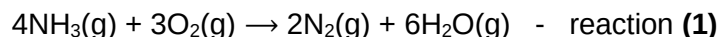
$$\text{Required ratio} = (2.0023 \times 10^{-3}) / (4.005 \times 10^{-3}) = \frac{1}{2}$$



[Total: 15]

- 2 Nitrogen is one of the most abundant elements on earth. It is present in many molecules that find applications in chemistry and biology, such as ammonia, haemoglobin and amino acids.

- (a) During the combustion of ammonia, it reacts to form nitrogen gas and water vapour as seen in the equation below:



- (i) State the role of ammonia in reaction (1). [1]

The standard entropies, $S^\ominus$, of the above species are tabulated.

	$S^\ominus / \text{J mol}^{-1} \text{K}^{-1}$
$\text{NH}_3(\text{g})$	192.8
$\text{O}_2(\text{g})$	205.2
$\text{N}_2(\text{g})$	191.6
$\text{H}_2\text{O}(\text{g})$	188.8

- (ii) The standard change of entropy of a reaction is the difference between the sum of the standard entropies of the products and the sum of the standard entropies of the reactants.

Calculate the standard entropy change of reaction (1). [1]

- (iii) Account for the sign in your answer to (ii). [1]

- (iv) The standard enthalpies of formation of ammonia and $\text{H}_2\text{O}(\text{g})$ are $-45.9 \text{ kJ mol}^{-1}$ and $-241.8 \text{ kJ mol}^{-1}$ respectively.

Calculate the standard enthalpy change of reaction (1). [1]

- (v) Hence, calculate the $\Delta G^\ominus$ of reaction (1). [1]

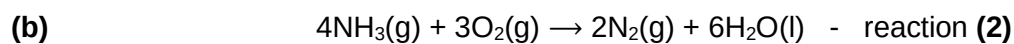
(i) Reducing agent / reductant

(ii) $2(191.6) + 6(188.8) - 4(192.8) - 3(205.2) = +129 \text{ J K}^{-1} \text{ mol}^{-1}$ (3 s.f.)

(iii) There is an increase in the amount of gas hence there is an increase in disorder.

(iv) $6(-241.8) - 4(-45.9) = -1270 \text{ kJ mol}^{-1}$ (3 s.f.)

(v) $-1267.2 - 298(129.2/1000) = -1310 \text{ kJ mol}^{-1}$ (3 s.f.)



The standard enthalpy change of reaction (2) is $-1531.2 \text{ kJ mol}^{-1}$.

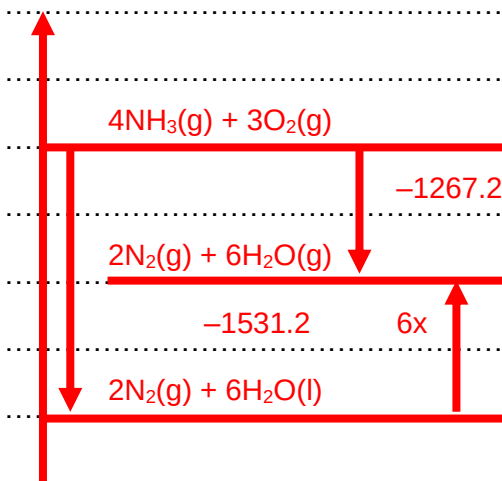
With the aid of an energy level diagram, find the standard enthalpy change of vaporisation of water. [2]

standard enthalpy change of vaporisation of water

$$= (-1267.2 + 1531.2) / 6$$

$$= +44.0 \text{ kJ mol}^{-1}$$

H / kJ mol^{-1}



[Total: 7]

- 3 Pentanoic acid is a carboxylic acid with molecular formula $C_5H_{10}O_2$.

Take note that your answers to (a), (b) and (c) should **NOT** be the same compound.

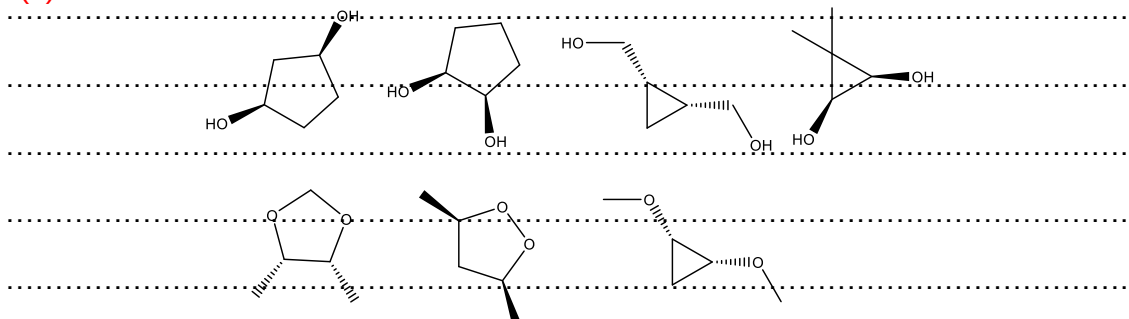
- (a) Draw the **stereochemical** formula of a constitutional isomer of $C_5H_{10}O_2$ which has chiral carbons but is achiral. [1]

Like other low-molecular-weight carboxylic acids, pentanoic acid has an unpleasant odour. In contrast, the esters of pentanoic acid tend to have pleasant odours and are used widely in perfumes and cosmetics.

- (b) Draw the **skeletal** formula of a chiral ester which is a constitutional isomer of $C_5H_{10}O_2$. [1]

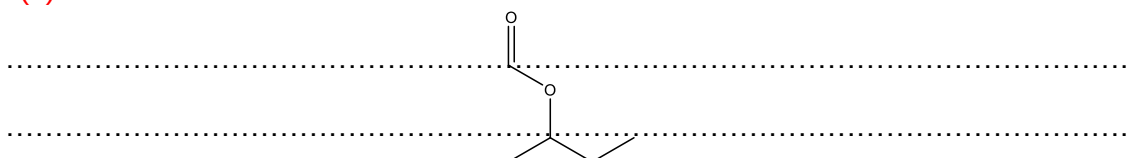
- (c) Draw the **skeletal** formula of a constitutional isomer of $C_5H_{10}O_2$ which does not show enantiomerism but shows cis-trans isomerism. [1]

(a)

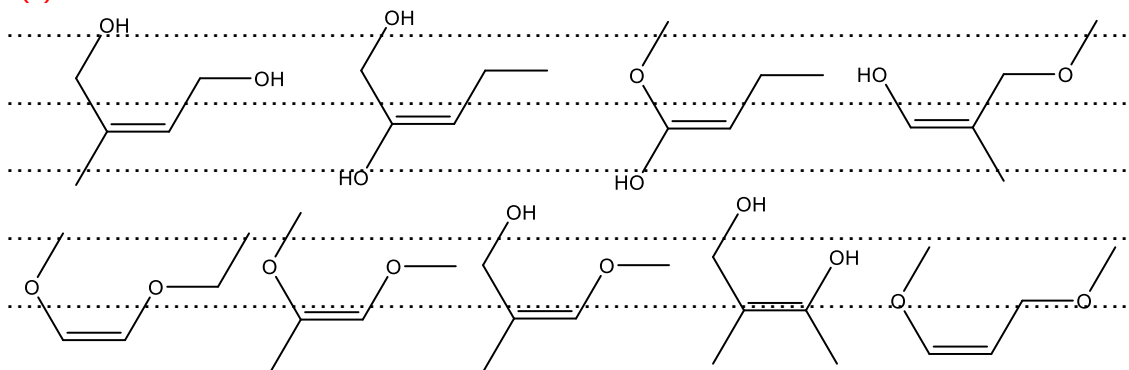


(Dash and wedges are MANDATORY.)

(b)



(c)



(or any of the structures in (a), as long as the structures proposed in (a) and (c) aren't the same.)

[Total: 3]

Additional answer space

If you use the following pages to complete the answer to any question, the question number must be clearly shown.

This image shows a full page of a document template designed for handwriting practice or general note-taking. It features a series of evenly spaced, horizontal dotted lines that run across the entire width of the page. The background is plain white, providing a clear contrast for the grey dots. There are no margins, headers, footers, or other markings present on the page.

